

THE SKY IS HAVING A BREAKDOWN

*400+ True Stories of Fire Tornadoes, Raining Fish,
Cloud Monsters, and the Strangest Weather Ever
Caught on Camera*

[Author Name]

Copyright © 2026 [Author Name]. All rights reserved.

No part of this book may be reproduced without written permission of the author, except for brief quotations in reviews.

This book contains facts about weather and natural phenomena drawn from published sources. It is for general interest and is not a safety manual. Never approach severe weather for photography.

First edition, 2026.

Contents

Introduction: Please Hold, the Sky Is Busy	4
Chapter 1: Clouds That Look Like Alien Invasions	6
Chapter 2: Hurricanes With Eyeballs	10
Chapter 3: Tornado Tales and Twisting Terrors	14
Chapter 4: Fire Tornadoes and Sky-High Infernos	18
Chapter 5: Snowflakes, Snow Donuts, and Self-Rolling Snowballs	22
Chapter 6: Ice That Grows Hair and Other Frozen Weirdness	26
Chapter 7: Lightning Strikes That Broke the Record Books	30
Chapter 8: Sprites, Elves, and Ball Lightning	34
Chapter 9: When It Rains Fish, Frogs, and Red Water	38
Chapter 10: Hailstones the Size of Grapefruits	42
Chapter 11: Walls of Dust and Sandstorms From Hell	45
Chapter 12: Rainbows, Moonbows, and Halos That Shouldn't Exist	49
Chapter 13: When the Sun Hits Earth With a Light Show	53
Chapter 14: Rain Records and Floods That Rewrote the Map	57
Chapter 15: Derechos, Downbursts, and Winds That Flatten Forests	61
Chapter 16: Scorching, Freezing, and Everything in Between	65
Chapter 17: Rogue Waves, Meteotsunamis, and Seas Gone Mad	68
Chapter 18: Animals Who Saw the Storm Coming	72
Chapter 19: The Daredevils and Gadgets That Caught It All	76
Chapter 20: Weather on Other Worlds	80
Sources and Further Reading	83

Introduction: Please Hold, the Sky Is Busy

In 1884, a photographer in Kansas pointed a camera at a tornado and made one of the earliest known photographs of one. Weather photography has been a risky hobby ever since. Cameras are cheaper now and they live in every pocket, car dashboard, doorbell, and satellite. Almost anything the atmosphere does gets recorded by somebody. The result is a visual record of a planet that regularly forgets it is supposed to be calm.

This book collects the strangest of it: hundreds of facts about rare clouds, hurricane eyes, fire whirls, snow that rolls itself into cylinders, lightning that crosses state lines, and storms on planets where the word "weather" gets a lot of extra zeros attached. Each chapter has a theme and between twenty and twenty-three facts. You can read straight through or open the book anywhere and start.

A few ground rules

Every fact in here is meant to be true. Where a number is a record, the book says who certified it. Where a record is disputed, you'll be told that too. If something is a theory, a legend, or an unverified claim, it carries a label:

- **[THEORY]** means scientists have a proposed explanation that isn't settled.
- **[MYTH]** means a popular belief the evidence doesn't support.
- **[UNCONFIRMED]** means it was reported, but nobody has verified it to the standard of a record.

- **[DISPUTED]** and **[DEBATED]** mean experts disagree about the measurement or the finding.
- **[UNOFFICIAL]** means a proposal that no governing body has adopted.

Weather records change. Researchers re-examine old storms and new instruments find old bolts. A famous example is lightning: in 2025 the World Meteorological Organization certified a flash from 2017 that nobody noticed at the time, because the data sat unexamined for years. Figures here are current as of writing. If a record matters to you, check the issuing agency.

Last thing. Please don't chase any of this on foot. The people who got these shots usually had training, equipment, and a good deal of luck. The facts will keep until you get indoors.

Chapter 1: Clouds That Look Like Alien Invasions

Clouds cover about two-thirds of the planet at any moment, and most of them behave. These ones don't. Some look like stormy oceans hung upside down, some punch holes in themselves, and one rolls across northern Australia at highway speed while glider pilots ride it for fun.

1. **A London pharmacist named every cloud.** In an 1802 lecture, Luke Howard proposed Latin names for cloud types: cumulus, stratus, cirrus, and combinations of them. His essay followed in 1803, and the names stuck. The German poet Goethe admired the system enough to write poems about Howard.
2. **A fair-weather cumulus weighs about as much as a hundred elephants.** NCAR scientist Peggy LeMone's estimate for a cloud roughly a kilometre across comes to around 500 metric tonnes of water droplets, or 1.1 million pounds. It stays up because the droplets are tiny and the air beneath is rising.
3. **The 2017 cloud atlas added a long list of new names.** The World Meteorological Organization's updated International Cloud Atlas included asperitas, volutus (roll cloud), fluctus (wave cloud), cavum (hole), murus (wall cloud), and cauda (tail cloud), among others.
4. **Some clouds are named for what made them.** The 2017 atlas also lists flammagenitus (born from fire), silvagenitus (born from forests), cataractagenitus (born from waterfalls), and homogenitus (born from human activity, such as the trails behind aircraft).

5. **Asperitas looks like a rough sea seen from below.** Its undulating, dark base prompted the Cloud Appreciation Society, founded in 2004 by Gavin Pretor-Pinney, to campaign for it to be recognised. The WMO added it in 2017. Despite its drama it doesn't usually bring severe weather, and meteorologists are still working out exactly how it forms.
6. **Kelvin-Helmholtz clouds are breaking waves in the sky.** They form where two layers of air slide past each other at different speeds, and they're named for the physicists Lord Kelvin and Hermann von Helmholtz. The curls often last only a minute or two before they smear out.
7. **The same instability shows up on other worlds.** Kelvin-Helmholtz waves have been observed along Saturn's magnetosphere and in the Sun's corona, as well as in Earth's own clouds.
8. **Lenticular clouds look like flying saucers, and have been reported as UFOs.** They form in the crest of a wave of air flowing over a mountain, which is why they hover above peaks like Mount Fuji and Mount Shasta without moving.
9. **That same mountain wave carried a glider to the stratosphere.** In 2006, Steve Fossett and Einar Enevoldson set a glider altitude record of 15,447 metres (50,671 feet) riding mountain wave over Argentina during the Perlan Project. The Perlan 2 glider later passed 76,000 feet in 2018.
10. **Mammatus clouds hang in pouches.** These bulges form on the underside of storm anvils where cooler, moister air sinks. They're associated with severe thunderstorms but they don't cause tornadoes, and they often show up after the worst has passed.

11. Hole-punch clouds have a cause scientists pinned down in 2011.

A study in *Science* by Andrew Heymsfield and colleagues showed fallstreak holes form when aircraft pass through a layer of supercooled water droplets. The droplets, below freezing and still liquid, abruptly freeze around the disturbance and fall out as ice, leaving a hole.

12. The hole often comes with a streak of ice crystals beneath it.

That streak is the ice that fell out of the cloud, evaporating before it reaches the ground. At sunset the whole thing can glow, and it has been mistaken for a UFO more than once.

13. The Morning Glory is a cloud that rolls. Near Burketown on the Gulf of Carpentaria in Queensland, long tubes of cloud can stretch across the sky, sometimes a few hundred kilometres long, and travel at speeds up to roughly 60 km/h. Glider pilots go there to surf the lifting air at their leading edge.

14. Nacreous clouds glow in pastel colours. Also called polar stratospheric clouds, they form at around 15 to 25 kilometres up, in extremely cold air in polar winter. Their pearly colours come from tiny ice particles diffracting sunlight, and they play a part in ozone destruction.

15. The highest clouds on Earth shine after dark. Noctilucent clouds sit about 76 to 85 kilometres up, high enough that the Sun still lights them when the ground below is dark. They were first documented in 1885. A popular idea says they appeared after the 1883 Krakatoa eruption, but that link remains **[THEORY]**.

16. Shelf clouds look like the lid of a spaceship. Also called arcus clouds, they form at the leading edge of outflow from a thunderstorm, when cool air pushes under warm, moist air and lifts it.

17. **Pileus clouds are caps on top of clouds.** A thin smooth cloud may form over a rising tower of cumulus as the tower pushes the air above it upward and cools it past the dew point. The cap can vanish as the tower punches through.
18. **Banner clouds trail from mountain summits.** On the downwind side of sharp peaks like the Matterhorn, air circles back and cools, forming a plume that looks like a flag attached to the summit even on an otherwise clear day.
19. **Clouds can make patterns that look like a river's wake.** Satellite images from Landsat show rows of alternating vortices, called Kármán vortex streets, streaming downwind of mountainous islands such as Chile's Juan Fernández Islands. The islands are small and the swirls are tens of kilometres across.
20. **A single lab session started cloud seeding.** On 13 November 1946, Vincent Schaefer of General Electric dropped dry ice into a cloud from a plane over Massachusetts and watched snow begin to fall. The experiment launched decades of weather modification attempts, whose effectiveness is still debated.
21. **Fog is a cloud that touches the ground, and some creatures drink it.** In the Namib Desert, darkling beetles of the genus *Stenocara* collect fog droplets on their backs and let the water run to their mouths. People now build mesh nets in places like Chile's Atacama to harvest fog the same way.

Chapter 2: Hurricanes With Eyeballs

A tropical cyclone is a heat engine the size of a small country. From a satellite it looks like a pinwheel with a hole in the middle, and the hole is the interesting part. These facts come from aircraft that flew into storms on purpose, buoys that rode them out, and cameras that kept rolling anyway.

1. **Hurricane Patricia (2015) had the strongest winds ever measured in a tropical cyclone.** The National Hurricane Center's post-season analysis put its peak at 215 mph (345 km/h) with a central pressure of 872 millibars, the lowest ever in the Western Hemisphere. That ties it with Typhoon Nancy of 1961 for the top wind speed on record.
2. **Typhoon Tip (1979) holds the lowest pressure ever recorded, at 870 millibars.** It was also the largest tropical cyclone on record, with a diameter of about 2,220 kilometres (1,380 miles), wider than the distance from Paris to Moscow.
3. **Patricia gained more than 100 mph of wind in about a day.** It was a tropical storm earlier that week, and its intensification was among the fastest forecasters have recorded. Then it hit a mountainous stretch of Mexico's Pacific coast and weakened quickly, which spared the country from the worst-case outcome.
4. **Eyewalls look like a stadium from inside.** Flight crews describe the "stadium effect": the eyewall curves outward as it rises, like seating tiers, while the clear eye sits at the bottom like a playing field. Some pilots report seeing blue sky overhead.

5. **Hurricane Wilma (2005) had an eye two nautical miles across.**
At its smallest, the eye was only about 2.3 miles wide, the smallest NOAA had ever recorded in an Atlantic hurricane. Wilma also hit 882 millibars, the lowest pressure measured in the Atlantic basin.
6. **Eyewalls replace themselves.** In an eyewall replacement cycle, an outer ring of thunderstorms forms around the original eyewall and slowly chokes it. The storm weakens, then often strengthens again with a wider eye. Observers first noticed it in aircraft and microwave satellite data.
7. **Hurricane Dorian (2019) nearly stopped moving over the Bahamas.** At its slowest, its forward speed dropped to about one mile per hour while it sat over Grand Bahama. It came ashore on Elbow Cay in the Abacos with sustained winds of 185 mph, tied for the strongest Atlantic landfall on record.
8. **The first deliberate flight into a hurricane was a bet.** On 27 July 1943, Colonel Joseph Duckworth flew a single-engine AT-6 Texan into a hurricane near Galveston, Texas, after British pilots told him it couldn't be done. He went back later that day with a weather officer, Lieutenant Ralph O'Hair.
9. **NOAA's hurricane planes have nicknames.** Two of its WP-3D Orion aircraft are called Kermit and Miss Piggy. In 1989, one of NOAA's P-3s lost an engine in the eyewall of Hurricane Hugo and the crew brought it home on the other three.
10. **A sailing drone filmed the inside of Hurricane Sam.** In September 2021, a Saildrone robotic vessel steered through the eyewall of Category 4 Hurricane Sam in the Atlantic. It measured waves of about 14 metres (46 feet) and gusts up to 126 mph, and it sent back the first video from the sea surface near a hurricane's eye.

11. **Typhoon Haiyan (2013) hit the Philippines with 195 mph winds.** That is the Joint Typhoon Warning Center's estimate of one-minute sustained speed at peak. Storm surge of several metres swept through Tacloban and the death toll topped 6,000.
12. **Hurricane Katrina's storm surge reached 27.8 feet.** That is the highest surge ever measured in the United States, recorded along the Mississippi coast in 2005. The surge, more than the wind, caused most of the deaths and destruction.
13. **The deepest known storm surge may be from 1899.** Cyclone Mahina struck Bathurst Bay, Australia, and researchers estimate a surge of around 13 metres (42 feet) based on evidence left on the land. It's an estimate from field evidence rather than an instrument reading, and some scientists have questioned the figure.
14. **One storm lasted a month.** Hurricane John in 1994 survived 31 days over the Pacific, covering roughly 13,000 kilometres (8,000 miles) and crossing the International Date Line twice. It remains the longest-lived tropical cyclone on record.
15. **Typhoon Vamei formed almost on the equator.** Tropical cyclones normally can't form within a few degrees of the equator because the Coriolis effect is too weak to start the spin. Vamei did it in December 2001, about 1.5° north of the line, near Singapore, and hit Malaysia.
16. **A hurricane showed up in the South Atlantic.** Many forecasters had considered that impossible until Catarina struck southern Brazil in March 2004, the first recorded hurricane there.
17. **Hurricane Ivan (2004) produced more than 100 tornadoes.** Outer bands of tropical cyclones often spin off short-lived tornadoes, and Ivan set records for the number it made as it moved inland over the southeastern United States.

18. **A hurricane's heat output is about 200 times the world's electrical generating capacity.** That's NOAA's figure for the heat energy a mature storm releases as water vapour condenses. Only a small fraction becomes wind.
19. **Two storms can orbit each other.** The Fujiwhara effect, described by the Japanese meteorologist Sakuhei Fujiwhara in a 1921 paper, happens when two cyclones come close enough to rotate around a shared centre. They can also merge.
20. **The Saffir-Simpson scale grew from an engineer's wind-damage chart.** Herbert Saffir, a structural engineer, made it around 1971, and the National Hurricane Center's director, Robert Simpson, added storm surge. The scale has five categories.
21. **There is no official Category 6.** Some scientists have proposed one for storms over 192 mph, since Patricia and a handful of typhoons have already exceeded that. This remains **[UNOFFICIAL]**, and the National Hurricane Center hasn't adopted it.
22. **Storms used to be named after politicians.** In the 1890s the Australian meteorologist Clement Wragge gave tropical storms the names of politicians he disliked. The United States started using women's names in 1953 and added men's names for Atlantic storms in 1979.
23. **Seabirds get stuck in the eye.** Sailors and aircraft crews have documented flocks of birds circling inside the calm centre, sometimes carried hundreds of kilometres from where they started.

Chapter 3: Tornado Tales and Twisting Terrors

Tornadoes are the smallest storms in this book and some of the hardest to measure. They last minutes, they destroy the instruments you point at them, and they've been caught on camera for 140 years. The footage taught researchers most of what they know.

1. **The widest tornado on record was 2.6 miles across.** The El Reno, Oklahoma tornado of 31 May 2013 spanned about 4.2 kilometres at its peak. It was rated EF3 based on damage, though radar measured winds in subvortices near 300 mph.
2. **The Tri-State Tornado (1925) stayed on the ground for 219 miles.** It moved at roughly 62 mph across Missouri, Illinois, and Indiana on 18 March 1925, killing 695 people. No continuous tornado track since has beaten it.
3. **The fastest tornado wind ever measured was 301 mph.** A Doppler on Wheels radar led by Joshua Wurman recorded it in the Bridge Creek–Moore tornado on 3 May 1999. The estimate carries an uncertainty of about 20 mph.
4. **April 1974 produced 148 tornadoes in under two days.** The Super Outbreak stretched across 13 states on 3 and 4 April and killed over 300 people. Thirty of its tornadoes were rated F4 or F5.
5. **The April 2011 outbreak produced over 360 tornadoes.** Across 25 to 28 April, the storms killed around 320 people, mostly in Alabama, and it remains the largest tornado outbreak in recorded history.

6. **The first tornado photographs date from 1884.** Two are among the oldest known: one from Kansas on 26 April 1884, near Garnett, and one from Howard, South Dakota on 28 August the same year. Exposure times were short enough, barely, to catch the funnels.
7. **The Fujita scale came from a man who also discovered downbursts.** Ted Fujita of the University of Chicago devised the F-scale in 1971. His later work identified microbursts, which changed how airports manage wind shear.
8. **No tornado has been rated EF5 since 2013.** The Enhanced Fujita scale replaced the original in 2007. The last EF5 was the Moore, Oklahoma tornado of 20 May 2013, and as of this writing none has been rated since.
9. **Hallam, Nebraska held the width record before El Reno.** The EF4 that struck on 22 May 2004 was about 2.5 miles wide at its largest. Its width was measured by survey, not radar.
10. **A probe once measured a 100-millibar pressure drop inside a tornado.** In June 2003, Tim Samaras placed a pressure probe in the path of a tornado near Manchester, South Dakota. It recorded the largest drop ever measured in a tornado, about 100 millibars.
11. **Tornadoes cross rivers.** The idea that rivers or hills protect a town is a **[MYTH]**. The Tri-State Tornado crossed the Mississippi, and tornadoes have run over hills, valleys, and downtown districts.
12. **Opening the windows to equalise pressure doesn't help.** That advice used to circulate widely and is a **[MYTH]**. Houses are damaged by wind and flying debris, not by pressure differences, and the time wasted could be spent getting to shelter.

13. **The deadliest tornado in history hit Bangladesh.** The Daulatpur–Saturia tornado of 26 April 1989 is estimated to have killed around 1,300 people. Dense population and weak housing contributed to the toll.
14. **The UK has more tornadoes per square mile than almost anywhere.** It records an average of about 30 to 50 a year, most of them weak. The United States, by comparison, sees roughly 1,200 annually.
15. **Waterspouts are common in the Florida Keys.** Fair-weather waterspouts there number in the hundreds each year, mostly weak, and form over warm shallow water under developing cumulus clouds.
16. **Some tornadoes contain several smaller vortices.** Multi-vortex tornadoes have suction vortices that spin around the main funnel like horses on a carousel. They explain why one house can be destroyed while its neighbour stands.
17. **Landspouts form without a thunderstorm's rotating updraft.** They grow from the ground up, when a rotating column of air under a growing cumulus is stretched upward. They look like tornadoes and are usually weaker.
18. **Gustnadoes are swirls of dust along a storm's gust front.** They look tornadic and sometimes have a funnel, but they aren't connected to a rotating cloud base above them.
19. **Tornadoes are most common in the United States.** The central plains see the clash of warm, moist Gulf air with dry air from the Rockies and cold air from Canada. Dixie Alley, across the Southeast, has fewer storms but more nighttime tornadoes and a higher death rate.

20. **A famous story says tornadoes can pluck chickens.** There are many reports across the twentieth century of birds found stripped of feathers after storms. Researchers have proposed that the force of wind and debris could do it, but individual cases are mostly anecdotal and **[UNCONFIRMED]**.
21. **Storm chasers have studied tornadoes since the 1970s.** The VORTEX2 project in 2009 and 2010 was the largest tornado research effort ever attempted, with dozens of vehicles and instruments deployed across the Plains.
22. **Tim Samaras died chasing.** Samaras, his son Paul, and Carl Young were killed at El Reno in 2013 when the tornado changed direction and size. The deaths are widely described as the first of storm chasers killed by a tornado, and they changed how many research teams plan their routes.

Chapter 4: Fire Tornadoes and Sky-High Infernos

Large fires make their own weather. They build clouds, pull in wind, and sometimes spin up vortices that rise a kilometre or more. Some of the events here caused terrible losses, and the facts are told plainly for that reason.

1. **The Carr Fire spawned a fire tornado rated EF3.** On 26 July 2018, near Redding, California, a fire whirl formed with winds that the National Weather Service estimated at over 140 mph. The Weather Service rated it after surveying the damage, an unusual step for a vortex that fire created.
2. **Canberra's 2003 firestorm made a confirmed fire tornado.** On 18 January 2003, a fire-induced vortex was identified by Australia's Bureau of Meteorology and rated F2. The fires killed four people and destroyed about 500 homes.
3. **A fire whirl killed roughly 38,000 people in 1923.** After the Great Kanto earthquake, fires swept Tokyo, and a firestorm swept through the Army Clothing Depot, where tens of thousands had taken shelter. It is one of the deadliest fire whirls in history.
4. **The Peshtigo Fire of 1871 killed more people than any other U.S. wildfire.** It burned across northeastern Wisconsin on 8 October, the same night as the Great Chicago Fire, and estimates put the death toll between 1,200 and 2,500. Survivors described winds and a "tornado of fire."

5. **Hamburg's 1943 firestorm pulled winds of hurricane force into the city.** The bombing raids of July 1943 created a firestorm that drew in air from all sides. Weather researchers have studied it since as an example of fire-driven wind.
6. **Pyrocumulonimbus clouds are thunderstorms built by fire.** Heat and smoke from a large fire rise into the atmosphere, pull in moisture, and form storm clouds that produce lightning, hail, and sometimes tornadoes. They're abbreviated pyroCb.
7. **An Australian smoke plume rose to the stratosphere.** After the Black Summer fires of 2019–20, a rotating smoke vortex reached about 35 kilometres (22 miles) up. Satellite instruments tracked the smoke around the Southern Hemisphere.
8. **Smoke can rise by heating itself.** Canadian wildfire smoke in 2017 reached the stratosphere and then climbed higher, absorbing sunlight and rising to around 23 kilometres, according to a 2019 study.
9. **PyroCb lightning can start new fires.** Lightning from fire-generated storms has ignited fresh blazes ahead of the original fire front, a hazard for firefighters because it spreads the fire beyond containment lines.
10. **Fire-born clouds got their own official name.** The 2017 International Cloud Atlas lists "flammagenitus" for clouds produced by fire, with pyrocumulonimbus as the extreme case.
11. **A blue whirl was discovered in a lab in 2016.** University of Maryland researchers burning fuel on water inside a fire-whirl chamber found that it could shift into a small, stable blue flame that burns very cleanly. They published it in *PNAS*.

12. **Fire whirls can occur in oil-spill burns.** The blue whirl experiment was done partly to study whether similar structures might burn spilled oil more efficiently. That use remains a research idea.
13. **Fire whirls come in all sizes.** Dust-devil-scale fire devils form on small grass fires all the time. The big ones, hundreds of metres tall and spinning for many minutes, are much rarer.
14. **Flames themselves can reach over 1,000°C.** Wildfire flame temperatures typically range from about 800 to 1,200°C, hot enough to melt aluminium. That's the temperature of the flame, not necessarily the heat felt downwind.
15. **A firefighter once saved himself by lighting another fire.** In the 1949 Mann Gulch fire in Montana, smokejumper foreman Wag Dodge lit an "escape fire" in the grass ahead of him and lay down in the burnt ground. He survived. Thirteen men died.
16. **New York's sky turned orange in 2023.** Smoke from Canadian wildfires drifted south, and on 7 June 2023 the city's air quality index passed 400 at times, among the worst measured in a major city that day.
17. **The smoke crossed the Atlantic.** Particles from those same Canadian fires were detected in Europe later in June 2023, including in Norway.
18. **California's sky went dark at noon in 2020.** On 9 September 2020, smoke over the San Francisco Bay Area turned daylight a deep orange, an event widely photographed and nicknamed Orange Day.

19. **Wildfires can reach cloud heights comparable to volcano eruptions.** The largest pyroCb events send material into the lower stratosphere, which is the same region large volcanic eruptions reach.
20. **Satellites see fires from orbit.** NASA's MODIS and VIIRS instruments detect thermal hot spots, and the International Space Station crew regularly photographs smoke plumes.

Chapter 5: Snowflakes, Snow Donuts, and Self-Rolling Snowballs

Snow is frozen water doing its best impression of lace. Under the right conditions it also rolls itself into logs, glows pink with algae, and falls on deserts. Here are twenty-two reasons to take a second look at the stuff.

1. **The first photograph of a snow crystal was taken in 1885.** A Vermont farmer named Wilson Bentley attached a bellows camera to a microscope on 15 January 1885 and got a clear picture. Over his life he photographed more than 5,000 snowflakes and published *Snow Crystals* in 1931. He died that December of pneumonia after walking home in a blizzard.
2. **Two snowflakes have been found that looked the same.** In 1988, Nancy Knight of the National Center for Atmospheric Research collected two simple hexagonal crystals in Wisconsin that appeared identical under the microscope. Complex branching flakes, which have far more ways to differ, almost certainly never repeat.
3. **A Japanese physicist grew the first artificial snow crystals in 1936.** Ukichiro Nakaya at Hokkaido University worked out that a crystal's shape depends on temperature and humidity at the time it grows. He is credited with the line "Snow crystals are letters sent from heaven."
4. **Snowflakes have six sides because water molecules arrange themselves into hexagons.** The hydrogen bonds between molecules lock into a six-fold lattice as ice forms. Dendrites, the classic feathery stars, grow fastest at around -15°C .

5. **Snow rollers form on their own.** When wet snow, a thin icy crust, and a steady wind line up, gusts push a clump across the field. It picks up layers like a rolled carpet. Photographs of fields scattered with them have circulated since the 1800s.
6. **Some snow rollers become donuts.** When the soft centre of a roller blows out or collapses, the log is left hollow. They are rare enough that finding one is a decent excuse to call a neighbour over.
7. **Thundersnow is real and unusually quiet.** Lightning during a snowstorm happens in strongly rising air, often near the Great Lakes or in intense winter cyclones. Snow muffles the thunder, so people close to the strike may hear only a dull boom.
8. **Penitentes are blades of snow, sometimes taller than a person.** Darwin described them in the Andes in 1835, and the name comes from their resemblance to hooded penitents kneeling in rows. They form when strong sunlight at high, dry altitude carves the snow into spikes pointing at the Sun.
9. **Pluto may have its own penitentes.** NASA's New Horizons probe photographed a region of bladed terrain in 2015 on a mountainous area of Pluto. Researchers proposed that the blades are methane ice shaped by the same process, though that is a **[THEORY]** based on modelling and remote imaging.
10. **Watermelon snow is red because of algae.** The species *Chlamydomonas nivalis* produces a red pigment that protects it from strong sunlight. The snow sometimes smells faintly sweet, and the dark colour makes the snow melt faster.
11. **Snow is mostly air.** Fresh snow is typically 90 percent air or more, which is why a foot of the stuff melts down to roughly an inch of water. The exact ratio varies from about 5:1 in wet snow to 20:1 or more in cold, dry powder.

12. **Mount Baker holds the single-season snowfall record.** The ski area in Washington State measured 1,140 inches (95 feet) in the 1998–99 winter, the most ever recorded for a season anywhere.
13. **Silver Lake, Colorado, got 75.8 inches of snow in 24 hours in 1921.** That's the accepted U.S. record for a single day, set on 14–15 April.
14. **An Italian town claims it beat that.** Capracotta reported about 256 centimetres (over 100 inches) in 18 hours in March 2015. The measurement was made by local observers and hasn't been formally verified as a world record, so treat it as **[UNCONFIRMED]**.
15. **The largest snowflake on record may not have been real.** A report from Fort Keogh, Montana, in January 1887 described flakes 15 inches wide and 8 inches thick. It's in record books, but nobody measured it with modern instruments, and many scientists doubt it.
16. **Snow squeaks when it gets cold.** Below roughly -10°C , packed snow underfoot makes a squeaking sound as ice crystals rub together. Warm snow is quieter and more slushy.
17. **Snow has fallen on the Sahara more than once.** Orange dunes in Aïn Séfra, Algeria, were dusted white in December 2016 and again in January 2018. Photos of the contrast spread across the internet within hours.
18. **The Atacama sees snow too.** One of the driest places on Earth received rare snowfall in 2011 and again in 2013, which stranded travellers and covered desert plants.

19. **There's snow on the equator.** Cayambe volcano in Ecuador sits almost exactly on the equator and its summit, above 5,700 metres, stays under permanent ice and snow. Hawaii's Mauna Kea is snow-capped most winters as well.
20. **A chinook once raised the temperature 49 degrees in two minutes.** On 22 January 1943, in Spearfish, South Dakota, the temperature jumped from -4°F to 45°F in about two minutes, a world record. Within hours it dropped back, cracking windows along the way.
21. **Diamond dust falls from a clear sky.** In very cold, still air, as in Antarctica and the Arctic, tiny ice crystals form near the ground and glitter in sunlight even when no cloud is overhead.
22. **Lake-effect snow can bury a city while the next suburb stays dry.** In November 2014, parts of the Buffalo, New York area got over five feet in three days from narrow bands off Lake Erie, while the city centre only a few miles north saw far less.

Chapter 6: Ice That Grows Hair and Other Frozen Weirdness

Ice does things you wouldn't expect a solid to do. It sprouts white filaments, makes spinning discs, sings, and builds spires underwater. These facts come from lakes, rivers, roads, and the polar seas.

1. Hair ice grows on dead branches, and a fungus is involved.

Thin silky strands, each a fraction of a millimetre across, extrude from the wood on cold, humid nights. In 2015, researchers Gerhard Wagner and Christian Mätzler identified the fungus *Exidiopsis effusa* as a key factor. Alfred Wegener, better known for continental drift, had proposed a biological cause in 1918.

2. Hair ice keeps its shape for hours. The fungus appears to slow the recrystallisation that would normally turn the fine hairs into lumps of ice. Remove the fungus from a branch and the effect doesn't appear.

3. Frost flowers bloom on new sea ice. On very cold, calm nights in polar regions, salty brine pushes up through a thin ice layer and freezes into feathery clusters. They're salt-rich and can host bacteria.

4. Brinicles are underwater icicles that freeze everything near them. Extremely cold, salty brine sinking from sea ice forms a tube that grows toward the seabed. A BBC film crew captured one in the Antarctic for *Frozen Planet* in 2011, and the brinicle froze starfish on the seabed in its path.

5. **Ice circles rotate on rivers.** In January 2019, a disc about 100 metres across was filmed turning slowly on the Presumpscot River in Westbrook, Maine. It made roughly one rotation every 30 minutes. Researchers think eddies in the current shear the edges of a floating chunk and keep it spinning.
6. **Ice volcanoes erupt without heat.** On the shore of Lake Michigan and other big lakes, waves surge through cracks in a shelf of ice and spray water upward. It freezes in a cone as it falls.
7. **Ice can shove itself ashore.** In Alaska, the Inuit word *ivu* refers to ice pushed up onto the land by wind or current. Piles can reach several metres high and have crushed buildings.
8. **Ice eggs wash up on beaches.** In November 2019, thousands of ice balls, some the size of footballs, covered a beach on Hailuoto island in Finland. They form when small pieces of slush tumble in the surf and gather layers of ice as they roll.
9. **Lake Baikal holds about 20 percent of the world's unfrozen surface fresh water.** It's also the deepest lake on Earth at around 1,640 metres. In winter its ice thickens enough for trucks to drive across, and the surface cracks with sounds like gunfire.
10. **Frozen methane bubbles stack like layered coins.** In some Baikal and Canadian lakes, bubbles rising from decaying matter below freeze in columns under clear ice. The bubbles look like white pancakes suspended in glass.
11. **Pancake ice is real and looks like it sounds.** In choppy polar seas, new ice forms round plates with raised rims because the plates bump into each other. They can be anywhere from a few centimetres to several metres across.

12. **Niagara Falls once stopped running.** On 29 March 1848, an ice jam at the eastern end of Lake Erie blocked the Niagara River upstream of the falls. For roughly 30 hours the flow slowed to a trickle, and people walked on the exposed riverbed.
13. **The 1998 North American ice storm brought down the power grid.** In January 1998, freezing rain coated Quebec, eastern Ontario, and parts of New York and New England. More than a million households in Quebec lost electricity, some for weeks, and steel transmission towers folded.
14. **Ice can sing.** Thermal contraction on a frozen lake sends waves through the ice that make laser-like zaps when you skip a stone or skate across. Recordings of it have been used in sound design for sci-fi films.
15. **The ground can explode in the cold.** A frost quake, or cryoseism, happens when water in soil freezes suddenly and the ground cracks with a boom. They hit parts of the U.S. Midwest during the January 2014 polar vortex.
16. **A glacier calving event was filmed in 2008.** On 28 May 2008 at Ilulissat Glacier in Greenland, photographers from the Extreme Ice Survey recorded what was then the largest calving event ever filmed. The ice face was about three miles wide and rose several hundred feet.
17. **An ice shelf the size of Rhode Island broke apart in weeks.** The Larsen B shelf in Antarctica shed about 3,250 square kilometres between late January and early March 2002. Satellites watched the pieces separate.

18. **Mercury has ice.** In 2012, NASA's MESSENGER probe found evidence of water ice in permanently shadowed craters near the planet's poles, despite daytime temperatures hot enough to melt lead.
19. **Ice can burn.** Methane hydrate, sometimes called fire ice, is a cage of water molecules trapped around methane. Hold a flame to a chunk and the gas ignites while the ice melts.
20. **Soap bubbles freeze into crystal globes.** At around -10°C to -15°C , a bubble blown outdoors can freeze in seconds, with crystals spreading across its surface. Frozen bubbles shatter easily and often collapse when they land.
21. **Snowball Earth is a hypothesis.** Some geologists argue that the planet was nearly or completely covered in ice about 700 million years ago. The idea is **[THEORY]**, still debated, particularly the question of whether the oceans were frozen all the way to the equator.
22. **The Mpemba effect says hot water can freeze faster than cold.** It's named after Erasto Mpemba, a Tanzanian student who noticed it in the 1960s. Experiments have given mixed results, and it remains **[DEBATED]**.

Chapter 7: Lightning Strikes That Broke the Record Books

Lightning is quick, so the records are measured in kilometres and fractions of a second. Most of what's here was found by satellite or high-speed camera. A few records were found years after the fact, in data nobody had looked at closely.

1. **The longest lightning flash ever recorded covered 829 kilometres.** The World Meteorological Organization certified it in 2025. It happened in October 2017, stretching from eastern Texas to near Kansas City. Nobody noticed at the time; researchers found it by re-examining satellite data.
2. **That is about the distance from Paris to Venice.** The flash beat the previous record by 61 kilometres, which was the 768-kilometre flash across Mississippi, Louisiana, and Texas on 29 April 2020.
3. **Before that, the record was 709 kilometres in Brazil.** It occurred on 31 October 2018 and held until the 2020 flash passed it.
4. **A single flash lasted more than 17 seconds.** On 18 June 2020, a flash over Uruguay and northern Argentina ran for 17.102 seconds, the longest-duration lightning flash the WMO has certified.
5. **Roy Sullivan was struck by lightning seven times.** The Virginia park ranger survived strikes between 1942 and 1977, and Guinness lists him as the person hit most often and surviving all of them.

6. **One lake in Venezuela has lightning on roughly 260 nights a year.** The Catatumbo lightning, over the mouth of the Catatumbo River at Lake Maracaibo, produces near-continuous storms in the evening. Satellite data show it among the most lightning-dense places on Earth. In early 2010 it paused for several weeks during a drought and then resumed.
7. **A bolt is hotter than the surface of the Sun.** The channel heats the air to roughly 30,000°C, about five times hotter than the Sun's surface, though the bolt is only a few centimetres wide.
8. **Earth gets around 40 to 50 lightning flashes every second.** That adds up to over a billion flashes a year, counting those inside clouds.
9. **Positive lightning is the "bolt from the blue."** These strikes carry a positive charge from the top of a storm cloud and can hit the ground tens of kilometres from the storm. They tend to be stronger and longer than ordinary negative strikes.
10. **High-speed cameras show a bolt building downward in steps.** B.F.J. Schonland's team in South Africa recorded the stepped leader in the 1930s using a special camera. Modern cameras run at hundreds of thousands of frames per second and show the leader branching toward the ground.
11. **The Empire State Building is hit about 20 to 25 times a year.** Tall structures act as lightning rods. The building's rod has done its job since 1931.
12. **A French experiment proved Franklin's idea in 1752.** Thomas-François Dalibard erected a tall iron rod at Marly-la-Ville and drew sparks from a passing storm on 10 May 1752, a month before Franklin's kite experiment.

13. **Lightning melts sand into glass tubes.** Fulgurites form when a strike fuses silica. Some run for more than a metre underground and branch like roots.
14. **Lichtenberg figures can appear on skin after a strike.** The red, fern-shaped marks, sometimes called lightning flowers, fade after hours or days and aren't burns.
15. **Thunder is a shockwave.** The superheated air expands faster than sound. Count the seconds between flash and bang and divide by three to estimate the distance in kilometres.
16. **Lightning kills about 20 people a year in the United States.** That's the recent annual average according to the National Weather Service, far lower than in past decades thanks to better shelter and warnings.
17. **Florida's Lightning Alley is the busiest in the U.S.** The corridor between Tampa and Orlando gets the highest density of strikes in the country because sea breezes from both coasts collide there each afternoon.
18. **Lightning helps fertilise the planet.** The heat of a bolt splits nitrogen molecules in the air, which form compounds that rain carries to the ground, where plants can use them.
19. **Satellites now watch lightning continuously.** NOAA's GOES-16, launched in 2016, carries the Geostationary Lightning Mapper, which records flashes across the Americas day and night. Much of the megaflick record work relied on it.
20. **A single bolt killed 21 people in 1975.** Guinness lists a strike on a hut in Manica, Zimbabwe on 23 December 1975 as the deadliest single lightning strike on record.

21. **Planes get struck about once a year.** Commercial aircraft are designed to carry the current around the passengers and out the wingtips and tail, and the strikes usually leave only small burn marks.

Chapter 8: Sprites, Elves, and Ball Lightning

The lightning most people see is the boring kind. Above the storm clouds, other discharges flash for a few thousandths of a second, in red, blue, and rings hundreds of kilometres wide. For decades pilots described them and scientists politely disagreed.

1. **The first picture of a sprite was an accident.** On 6 July 1989, John Winckler of the University of Minnesota was testing a low-light camera for a rocket experiment when it recorded a brief red flash above a distant storm in the frame. It was the first image of a red sprite.
2. **Sprites flash 50 to 90 kilometres up and last a few milliseconds.** They appear above active thunderstorms as red, tentacled columns. Pilots had reported seeing them for decades before cameras caught them.
3. **The name came from folklore.** Researchers at the University of Alaska Fairbanks coined "sprite" in the early 1990s for something that was easy to miss and hard to prove existed.
4. **Blue jets shoot up from the tops of storms.** They climb at roughly 100 kilometres per second to about 40 to 50 kilometres in altitude. They were first recorded from aircraft in 1994.
5. **Gigantic jets reach the ionosphere.** In 2001, a team recorded a jet from a thunderstorm near Puerto Rico that extended roughly 70 kilometres upward and connected the storm to the ionosphere. A study of it appeared in *Nature* in 2002.

6. **ELVES are expanding rings of light.** The name stands for Emission of Light and Very Low Frequency perturbations from Electromagnetic Pulse Sources. They appear at about 90 kilometres and can spread out to roughly 400 kilometres across in a fraction of a millisecond. Japanese researchers recorded them in the 1990s.
7. **A camera on the space station watches for them.** The European Space Agency's ASIM instrument arrived on the ISS in April 2018 to study upper-atmosphere lightning and the gamma rays that sometimes come with it.
8. **Thunderstorms make gamma rays.** Terrestrial gamma-ray flashes were discovered in 1994 by NASA's Compton Gamma Ray Observatory, which was built to look at the cosmos but kept getting hit from below.
9. **They also make antimatter.** NASA's Fermi telescope detected positrons, the antimatter version of electrons, coming from terrestrial gamma-ray flashes in 2011. The finding was published by a team led by Michael Briggs.
10. **Ball lightning was recorded with a spectrograph for the first time in 2012.** A research team on the Qinghai Plateau in China was filming ordinary lightning when a luminous ball appeared after a strike. It lasted about 1.6 seconds, and the spectrum showed silicon, iron, and calcium, elements found in soil. The 2014 paper concluded the ball came from vaporised soil, one **[THEORY]** among many.

11. **Ball lightning has been reported for centuries and is still poorly understood.** Eyewitness accounts describe glowing spheres drifting through rooms or along the ground. Scientists don't agree on a single mechanism, and few events have been recorded by instruments.
12. **A 1638 church storm in Devon may involve ball lightning.** Accounts of the Great Thunderstorm at Widecombe-in-the-Moor on 21 October 1638 describe a fireball entering the church and killing several worshippers. The details come from pamphlets of the time, and the identification as ball lightning is **[UNCONFIRMED]**.
13. **A scientist may have died studying lightning in 1753.** Georg Wilhelm Richmann was killed in St. Petersburg in August 1753 while experimenting with atmospheric electricity. Some accounts describe a ball of fire. The details are secondhand and **[UNCONFIRMED]**.
14. **Volcanoes make lightning.** Ash particles rub together in an eruption plume and build up charge. The "dirty thunderstorm" effect has been photographed at Sakurajima in Japan, Taal in the Philippines, and Iceland's Eyjafjallajökull in 2010.
15. **The Hunga Tonga eruption of January 2022 set the lightning-rate record.** On 15 January, the plume produced about 192,000 flashes over roughly 11 hours, peaking at around 2,600 flashes a minute. Some of the lightning occurred 20 to 30 kilometres up, higher than any seen before.
16. **St. Elmo's fire is a glow, not a flame.** It's a corona discharge: electric fields on a ship's mast or an aircraft windscreen ionise nearby air. Sailors named it after Saint Erasmus, nicknamed Elmo, the patron saint of mariners.

17. **Dry lightning starts fires.** In mid-August 2020, thunderstorms over California delivered more than 10,000 strikes in about three days with little rain. They ignited hundreds of fires, including the large complexes that burned for weeks.
18. **"Heat lightning" is a [MYTH].** The flashes people see on warm nights are ordinary lightning from storms too far away to hear. Thunder usually can't be heard beyond about 15 miles.
19. **Whistlers are radio waves from lightning that travel into space.** They follow Earth's magnetic field lines and return as a falling whistle. Operators on long telephone lines heard them in the nineteenth century, and soldiers heard them on field phones in the First World War.
20. **Lightning keeps the planet humming at 7.83 hertz.** The Schumann resonances, predicted by Winfried Otto Schumann in 1952, are standing waves in the space between the ground and the ionosphere, excited by lightning strikes worldwide.

Chapter 9: When It Rains Fish, Frogs, and Red Water

People have been reporting animals falling out of the sky for at least two thousand years. Some of these stories are legend. A few are documented well enough that scientists take them seriously. This chapter tries to keep the two apart.

1. **Pliny the Elder catalogued strange rains.** His *Natural History*, finished around 77 CE, lists showers of milk, blood, flesh, iron, and wool among the portents reported in the Roman world. They're secondhand accounts and can't be checked.
2. **Athenaeus quoted older writers on a fish rain.** In the *Deipnosophistae*, written around 200 CE, he cites earlier authors describing fish that fell on parts of the Balkans, with stories of days of fish. It's among the oldest written reports of animals falling from the sky.
3. **A U.S. biologist saw it happen in 1947.** On 23 October 1947, in Marksville, Louisiana, A.D. Bajkov of the state's Department of Wildlife and Fisheries reported fish dropping onto the town during a morning of light rain. He published an account in *Science* in 1949.
4. **The usual explanation is a waterspout.** The idea is that a strong updraft over water could lift small fish or frogs and drop them miles away. It's a **[THEORY]**: reports of fish rains fit it, but nobody has directly watched a waterspout carry animals.

5. **Fish rains tend to feature one species.** Many accounts say all the fallen animals were the same kind and about the same size. That pattern is what you would expect from a vortex that sorts by weight, and it's one reason the waterspout explanation has lasted.
6. **Lajamanu, Australia, has had fish rain.** In February 2010, residents of this Northern Territory community reported hundreds of small spangled perch falling after a storm, in an area far from open water. Locals said it had also happened in 2004.
7. **In Yoro, Honduras, people say fish fall every year.** The *Lluvia de Peces* festival marks an event residents describe as happening between May and July after heavy storms. Scientists haven't confirmed the cause, and the story rests largely on local testimony.
8. **Frog rains turn up in newspaper archives.** Reports of frogs falling during storms appear in many countries and centuries. Few were examined by scientists at the time, so most remain **[UNCONFIRMED]**.
9. **In 1876, meat fell in Kentucky.** Chunks of a fleshy substance fell near Olympian Springs, Bath County, from a clear sky. Examiners at the time described them as lung tissue and cartilage. One later suggestion is that vultures vomited as they flew, which remains a **[THEORY]**.
10. **Red rain fell in Kerala in 2001.** After a loud sound heard in the area, red-coloured rain fell for several weeks in parts of southern India. Physicist Godfrey Louis proposed that the particles were extraterrestrial cells, which is a **[THEORY]** that mainstream scientists reject. An Indian government study identified them as spores of a common terrestrial algae, *Trentepohlia*.

11. **Saharan dust can colour rain red.** In Europe, storms that mix with dust from North Africa sometimes leave cars and windows coated in orange-red mud. It's often called "blood rain" in the press.
12. **Raindrops aren't teardrop-shaped.** A falling drop flattens at the bottom into a shape closer to a hamburger bun. Drops larger than about 6 millimetres break up before they reach the ground.
13. **The smell of rain has a name.** In 1964, two Australian scientists, Isabel Bear and Richard Thomas, called it petrichor in a paper in *Nature*. Humans can detect geosmin, one of the compounds responsible, at a few parts per trillion.
14. **High-speed cameras caught rain releasing the smell.** In 2015, MIT researchers filmed raindrops hitting porous surfaces and showed that tiny bubbles in the drop burst upward and shoot aerosols into the air, carrying scents from the soil.
15. **Virga is rain that never lands.** The streaks hanging under a cloud are precipitation evaporating in dry air before it reaches the ground. In deserts, a storm can look like it's pouring over a landscape that stays bone dry.
16. **Hail has fallen out of cloudless skies in some reports.** Ice chunks the size of fists have been reported falling from clear weather, including in Spain in 2000. Researchers called them megacryometeors, and their origin is still unexplained, with ideas ranging from aircraft ice to unusual atmospheric processes. Treat it as [UNCONFIRMED].
17. **Snow can be red and pink.** Alpine and polar snow sometimes turns red, orange, or pink because of algae living in it. The "watermelon snow" in Chapter 5 is the best-known example.

18. Rain on the equator may be more reliable than anywhere else.

The Intertropical Convergence Zone, a belt of rising air near the equator, brings near-daily storms in some regions. Its position shifts north and south through the year, and the monsoon follows.

19. "Raining cats and dogs" is not about animals falling.

The origin of the expression is unclear, with several proposed explanations and no firm evidence for any of them. Treat all of them as **[THEORY]**.

20. Ordinary rain is slightly acidic.

Carbon dioxide dissolving in raindrops gives clean rain a pH of around 5.6. The term "acid rain" was coined in 1872 by the Scottish chemist Robert Angus Smith, after he measured polluted rain in industrial Manchester.

Chapter 10: Hailstones the Size of Grapefruits

Hail forms when ice gets tossed up and down inside a storm, adding a layer each time. The bigger the updraft, the longer the ice stays aloft and the larger it grows. On the Plains and in northern Italy, that leads to some very large rocks.

1. **The biggest hailstone in U.S. history fell in Vivian, South Dakota.** On 23 July 2010 a stone measuring 8.0 inches (20.3 cm) across fell and weighed 1.94 pounds. Both numbers are U.S. records, confirmed by the National Climatic Extremes Committee.
2. **Europe's record is a 19-centimetre stone from Italy.** It fell at Azzano Decimo on 24 July 2023, breaking a record set only five days earlier when a 16-centimetre stone landed at Carmignano di Brenta.
3. **The heaviest stones on record fell in Bangladesh.** The World Meteorological Organization recognises stones of up to 1.02 kilograms (2.25 pounds) at Gopalganj on 14 April 1986. The storm is reported to have killed 92 people.
4. **Moradabad, India, had a deadly hailstorm in 1888.** On 30 April, stones reported to be as large as oranges fell during a severe storm. Records list more than 200 deaths, though the figures come from historical accounts.
5. **A Nebraska stone held the diameter record for seven years.** The hailstone at Aurora, Nebraska, on 22 June 2003 measured 7 inches (17.8 cm) across and held the U.S. record until Vivian beat it in 2010.

6. **Hailstones have rings like trees.** Cut one open and you'll see alternating clear and cloudy layers. Researchers read them as a record of the stone's trips through different parts of the storm.
7. **Big stones fall fast.** Baseball-sized hail drops at around 100 miles an hour. That speed, not the weight alone, is why cars and roofs take such punishment.
8. **Calgary's 2020 hailstorm was the costliest in Canadian history.** The storm on 13 June 2020 caused insured losses of about C\$1.2 billion, mostly to cars and roofs in the city's northeast.
9. **Denver's hailstorm of 1990 caused heavy insured damage too.** The storm on 11 July 1990 produced losses of about \$625 million, making it one of the costliest in the United States at that time.
10. **Hail alley covers the High Plains.** The corner where Colorado, Wyoming, and Nebraska meet sees more hail days than almost anywhere in North America because the storms form where cold air aloft meets moist, hot air at the surface.
11. **Guadalajara was buried in hail in June 2019.** An intense storm on 30 June dropped hail that piled up in drifts reported as deep as 1.5 metres in parts of the city. Residents shovelled it off their streets as if it were snow, and the hot-weather city photographed cars half buried.
12. **Hail grows in a cold band of the storm.** The stone gains most of its mass between roughly -10°C and -30°C , where supercooled droplets freeze on contact and add a layer.
13. **A supercell's updraft can exceed 100 miles an hour.** Meteorologists estimate that the largest stones need updrafts of about that speed or more to stay aloft long enough to reach their size.

14. **Hail often falls in narrow swaths.** A storm may flatten crops along a strip a mile wide and leave the next farm untouched. Insurance adjusters and weather services map them with radar and ground reports.
15. **Radar can estimate hail size, with limits.** Dual-polarisation radar measures the shapes and sizes of particles. It tells forecasters where the large stones likely are, but measuring in the field remains the best check.
16. **Hail is not frozen rain.** Frozen rain is sleet, small clear pellets. Hail forms only in thunderstorms and grows from layering. Graupel, a softer snowy pellet, is its smaller relative.
17. **Hail can be piled up by rain and wind.** A mass of melting hailstones can look like snow cover, and in extreme cases flows like slush into drainage systems, blocking them and causing flash flooding.
18. **Hail suppression is controversial.** Programs in places like Russia, Canada, and France have seeded storms with silver iodide to try to reduce hail. Scientists still debate whether it works, and the evidence is mixed, so any claim about effectiveness remains **[UNCONFIRMED]**.
19. **Europe's hail season is growing worse.** European researchers have reported rising hail damage trends in recent years, with 2023 setting multiple records for stone size. Scientists are still working out how much of that is climate change and how much is natural variability.
20. **Insurers treat hail as a major peril.** In the United States, hail causes billions of dollars in damage to roofs and vehicles in an average year, and some regions require special roofing material.

Chapter 11: Walls of Dust and Sandstorms From Hell

A thunderstorm's cold downdraft can scoop up desert soil and push it ahead as a wall. A dry year on a plowed prairie can send the entire topsoil east. And the Sahara quietly feeds the Amazon.

1. **"Haboob" comes from Arabic.** The word comes from *habūb*, meaning blasting or drifting, and describes a dust wall pushed out by a thunderstorm. It's most associated with Sudan, Arabia, and the American Southwest.
2. **A dust wall over Phoenix on 5 July 2011 rose over a mile high.** The storm rolled across the city in the evening, cut visibility to near zero, and cut power to thousands. Photos and time-lapse videos spread worldwide.
3. **A haboob is a thunderstorm's cold outflow.** When rain falls into dry air and evaporates, the air cools and sinks, then spreads out along the ground as a gust front carrying dust before it.
4. **Black Sunday hit the Plains on 14 April 1935.** A huge dust storm swept across Oklahoma, Texas, Kansas, and neighbouring states in the afternoon. Witnesses described sudden darkness and static electricity strong enough to stall cars.
5. **A reporter named the Dust Bowl the day after.** Robert Geiger of the Associated Press used the phrase "Dust Bowl" in a dispatch after Black Sunday, and it stuck as the name for the region and the era.

6. **Dust from the Plains fell on Washington.** In May 1934, a large storm carried topsoil east. Dust fell on cities from Chicago to New York and was reported on ships hundreds of miles out in the Atlantic.
7. **The Sahara fertilises the Amazon.** NASA satellite data show about 182 million tons of dust leaving the Sahara in a year, with about 27.7 million tons settling over the Amazon basin. It brings roughly 22,000 tons of phosphorus, a nutrient that rain forest soils lose.
8. **The Bodélé Depression in Chad is the planet's biggest single dust source.** The dry lake bed, the remains of ancient Lake Mega-Chad, supplies a large share of the dust that travels west.
9. **"Godzilla" crossed the Atlantic in June 2020.** An exceptional Saharan dust plume reached the Caribbean and the southern United States. Air quality in parts of the Caribbean was described as the worst in decades.
10. **The 2018 Martian dust storm silenced a rover.** A dust storm that grew to cover the whole planet in mid-2018 blocked sunlight from the solar panels of NASA's Opportunity rover. Contact was lost on 10 June 2018, and NASA declared the mission over in February 2019.
11. **Dust devils clean Martian solar panels.** Mars rovers Spirit and Opportunity lasted longer than planned partly because wind events, including dust devils, blew dust off their panels. Cameras have photographed dust devils crossing the Martian surface many times.

12. **Perseverance filmed dust devils too.** The rover's cameras and weather sensors have recorded them passing near Jezero Crater, giving scientists data on their size and the pressure drops they cause.
13. **Dust darkens mountain snow.** In Colorado's Rocky Mountains, dust blown from desert regions has been shown to speed up spring snowmelt by a few weeks, which affects the river flows downstream.
14. **Valley fever travels in the dust.** The fungus *Coccidioides* lives in soils of the American Southwest, and storms can lift its spores. Public health officials in Arizona and California warn about it after major dust events.
15. **Beijing's worst storm in a decade came in March 2021.** Dust from the Gobi Desert turned the city's sky orange and pushed air quality readings beyond the top of standard scales.
16. **Iraq endured a string of dust storms in spring 2022.** Several events in April and May sent thousands of people to hospital with breathing problems and closed airports. Drought and land degradation have made them more frequent.
17. **Shrinking lakes feed the dust.** The Aral Sea in Central Asia lost most of its water in the twentieth century, leaving a dry bed whose salty dust is carried by wind. Owens Lake in California was a similar source before the city of Los Angeles began controlling it.
18. **Dust storms cause highway pile-ups.** Sudden zero visibility on interstates in Arizona, New Mexico, and Texas has caused multi-vehicle crashes. Transportation departments there advise drivers to pull off the road and turn off their lights.

19. **A tall dust devil can reach a kilometre or more.** On Earth, they usually last minutes. On Mars, some towers rise to several kilometres.
20. **Sand dunes can sing.** Some dune faces avalanche and make a low booming sound audible from far away. Marco Polo wrote of the "voices of spirits" in the deserts of Central Asia, and the effect is now studied by physicists, though the exact mechanism is still debated.

Chapter 12: Rainbows, Moonbows, and Halos That Shouldn't Exist

Light bends, splits, and bounces around inside drops of water and crystals of ice. Almost every strange glow in the sky can be explained by the geometry of that process. Some are still rare enough to make people pull over.

1. **A primary rainbow appears 42 degrees from the point opposite the Sun.** The secondary bow sits near 51 degrees, with its colours reversed. Between the two lies a darker patch of sky.
2. **That dark strip has a name.** It's called Alexander's dark band, after Alexander of Aphrodisias, a Greek philosopher who described it around 200 CE.
3. **Rainbows are full circles.** The ground gets in the way. Passengers in aircraft, and hikers on peaks above the cloud, sometimes see the whole ring.
4. **Extra stripes inside a bow are called supernumerary arcs.** The faint pink-and-green bands appear when light waves from different paths interfere. Thomas Young used them in the early 1800s as evidence that light behaves as a wave.
5. **Twinned rainbows start from one point and split.** The cause is a mix of drop sizes in the rain, and photographs of them are rare.
6. **A moonbow needs a bright full Moon and a dark sky.** They're thousands of times fainter than a rainbow, so the eye sees them as a pale white arc. Cameras with long exposures show colours. Victoria Falls, Yosemite, and Cumberland Falls in Kentucky are known for them.

7. **Fogbows are nearly white.** The droplets in fog are so small, under about 0.05 millimetres, that light diffracts and the colours overlap. Fogbows look like a broad ghostly band.
8. **A rainbow in Taiwan lasted almost nine hours.** Guinness recognises the longest-lasting rainbow as 8 hours and 58 minutes, observed on Yangmingshan, Taipei, on 30 November 2017.
9. **"Double Rainbow Guy" made one go viral.** Paul Vasquez posted a video of a double rainbow near Yosemite in 2010 and it became a meme within weeks.
10. **A circumhorizontal arc looks like a rainbow lying on its side.** Often called a "fire rainbow," it's not a rainbow at all. It forms when sunlight passes through flat, hexagonal ice crystals in high cirrus, and the Sun must be more than about 58 degrees above the horizon.
11. **Sun dogs are bright spots 22 degrees from the Sun.** They form when plate-shaped ice crystals drift with their faces horizontal. They sit on both sides of the Sun, and are called parhelia in the technical literature.
12. **A painting from 1535 records a halo display.** The Vädersolstavlan, or "Sun Dog Painting," in Stockholm's Storkyrkan church depicts an elaborate halo display seen over the city that year. The version on display is a later copy of the original.
13. **A halo around the Sun or Moon is a 22-degree ring.** Hexagonal ice crystals in thin cirrus clouds bend the light. Folklore says a lunar halo means rain is coming, and thin cirrus can be a sign of an approaching front, though the link is unreliable.

14. **Light pillars rise from city lights on cold nights.** Falling plate crystals reflect lamps and headlights into vertical columns. They appear in the Arctic and Canada and are common in photographs of snowy towns.
15. **The green flash lasts a second or two.** At sunset or sunrise over a clear horizon, the atmosphere bends the sun's blue-green light slightly above the red. It's most reliably seen over the sea.
16. **A Brocken spectre is your own shadow.** Named for the Brocken mountain in Germany's Harz range, where Johann Silberschlag described it in 1780, it's your shadow thrown onto cloud below and often surrounded by a rainbow-coloured ring called a glory.
17. **Fata Morgana looks like floating castles.** This layered mirage forms when a temperature inversion bends light along the horizon. It's named after Morgan le Fay and is often reported in the Strait of Messina.
18. **A mirage may have fooled Arctic explorers.** In 1906, Robert Peary reported seeing land to the northwest, which he named Crocker Land. A 1913 expedition went looking and found nothing. Many historians think it was a mirage, though some dispute that, so treat the identification as **[THEORY]**.
19. **The Novaya Zemlya effect makes the Sun appear when it shouldn't.** In 1597, the crew of explorer Willem Barentsz saw the Sun return about two weeks early after the polar night. The cause is strong atmospheric refraction over very cold air.
20. **Sunset is red because blue light scatters away.** Lord Rayleigh worked out the scattering in the 1870s. At low sun, light travels through more air, so the blues scatter out of the beam and the reds stay.

21. **Sun rays seem to fan out, but they're parallel.** Crepuscular rays look like they radiate from the Sun because of perspective, the same reason railway tracks seem to converge. Sometimes they reappear on the opposite horizon as anticrepuscular rays.

Chapter 13: When the Sun Hits Earth With a Light Show

Space weather isn't weather in the usual sense, but it hits the same instruments, and people point their cameras at it. When the Sun throws a large burst of plasma at Earth, the sky lights up and some of our hardware fails.

1. The Carrington Event of 1859 is the strongest storm on record.

On 1 September, astronomer Richard Carrington saw a white-light flare on the Sun. A geomagnetic storm arrived within about a day, and auroras were reported as far south as the Caribbean.

2. Telegraphs sparked and some kept running with no power.

Operators in North America and Europe reported shocks and fires at their equipment. Some sent messages for a while using only the current induced by the storm.

3. A 1989 storm blacked out Quebec. On 13 March, a geomagnetic storm collapsed Hydro-Québec's grid in about 90 seconds. Six million people were without power for around nine hours.

4. The Halloween storms of 2003 darkened part of Sweden. In late October, a series of strong flares and ejections caused a blackout in Malmö and damaged or disrupted satellites. Airlines rerouted polar flights.

5. May 2024 brought the strongest storm in two decades. On 10 and 11 May, a G5 geomagnetic storm, the top of NOAA's scale and the first since 2003, lit auroras as far south as Florida and Mexico. Some farm GPS systems in the U.S. Midwest were disrupted.

6. **The Sun's activity cycle peaked around 2024.** NASA and NOAA said in October 2024 that the Sun had reached the maximum period of Solar Cycle 25. Aurora sightings at low latitudes rose along with it.
7. **An X28 flare in 2003 may be the largest ever measured.** The flare on 4 November 2003 saturated the detectors, so its strength is an estimate, about X28.
8. **A near miss happened in 2012.** On 23 July 2012, a massive coronal mass ejection crossed Earth's orbit nine days before Earth got there. NASA's STEREO-A spacecraft recorded it. Researchers estimate it was comparable to the Carrington Event.
9. **A magnetic storm killed Starlink satellites in 2022.** SpaceX launched 49 satellites on 3 February 2022, and a geomagnetic storm increased drag in the upper atmosphere. About 38 of them were lost.
10. **A 1972 storm set off U.S. Navy sea mines.** On 4 August 1972, magnetic mines near Hải Phòng in North Vietnam detonated unexpectedly. Researchers later connected the explosions to a large geomagnetic storm.
11. **Green aurora comes from oxygen about 100 to 150 kilometres up.** Red appears higher, above about 200 kilometres, and blue or purple edges come from nitrogen.
12. **Aurora hunters named a new light "Steve."** Photographers in Alberta noticed a narrow purple ribbon in 2016 and joked about the name, borrowed from the film *Over the Hedge*. Scientists later turned it into a backronym: Strong Thermal Emission Velocity Enhancement.

13. **STEVE sometimes comes with a "picket fence."** The green streaks that hang beneath the purple band were identified in 2018 as a separate effect, though what produces them is still being studied.
14. **Aurora might make sound.** In 2016, Finnish researcher Unto Laine's team recorded crackling noises that appeared linked to auroral activity, apparently generated about 70 metres above the ground near a temperature inversion. The proposed mechanism is a **[THEORY]**.
15. **Auroras happen on other planets.** Hubble and the Juno spacecraft have imaged powerful auroras at Jupiter's poles, and Saturn has its own.
16. **Astronauts look down on the aurora.** The International Space Station orbits at about 400 kilometres, above the glow, which sits roughly 100 to 300 kilometres up. Crews photograph the curtains from above, sometimes with the station flying through the edge of them.
17. **Aurora can be seen from both poles at the same time.** The Northern and Southern lights often mirror one another in shape. Satellites have imaged both ovals at once.
18. **Solar wind travels at about 400 kilometres a second.** During storms it can be double that or more. A fast coronal mass ejection can cross the distance to Earth in less than a day.
19. **Citizen scientists track storms.** Projects like Aurorasaurus ask people to report sightings, and amateur photos helped identify STEVE.

20. **Satellite operators raise orbits during storms.** Heating of the upper atmosphere makes it expand, increasing drag on low satellites. Agencies issue warnings on NOAA's five-point scales for geomagnetic, solar radiation, and radio blackout storms.

Chapter 14: Rain Records and Floods That Rewrote the Map

An inch of rain on a lawn is a nuisance. Eighty inches on a mountain is a record. Flash floods turn dry streets into rivers in minutes, and the footage is often shot by people standing too close.

1. **Foc-Foc, Réunion, holds the 24-hour rainfall record: 1,825 millimetres.** The WMO recognises the reading from 7 and 8 January 1966, during Tropical Cyclone Denise. That's about 72 inches in a day, on an island with mountains over 3,000 metres high that force the moist air upward.
2. **Cherrapunji set the 12-month record in the 1860s.** The Indian hill town recorded about 26,470 millimetres between August 1860 and July 1861, which is more than 86 feet of rain.
3. **Mawsynram and Cherrapunji argue over "wettest."** Both lie in Meghalaya, India, and they trade the title depending on how the average is calculated. Both receive well over 10,000 millimetres in an average year.
4. **A minute of rain in Maryland set a record.** At Unionville on 4 July 1956, 31.2 millimetres (1.23 inches) fell in one minute, the highest rate recognised by the WMO.
5. **Hurricane Harvey dropped over 60 inches on parts of Texas.** In August 2017, the storm stalled over southeastern Texas and produced the highest rainfall total from a tropical cyclone in the continental United States.

6. **Zhengzhou received 201.9 millimetres in a single hour.** On 20 July 2021, the Chinese city recorded the highest hourly figure since measurements began in 1951. Fourteen people died when subway Line 5 flooded, and the province's death toll was reported in the hundreds.
7. **A Colorado canyon flood killed 144 people.** On 31 July 1976, about 12 inches of rain fell in roughly four hours over the Big Thompson River, and a wall of water swept down the canyon during a holiday weekend.
8. **The United Arab Emirates had its heaviest rain since records began in 1949.** On 16 April 2024, a persistent storm system dropped 254.8 millimetres in 24 hours at Khatm Al Shakla, near Al Ain, and 144 millimetres at Dubai International Airport. Cloud seeding was rumoured as a cause, but experts pointed to the storm system itself and warm seas.
9. **Pakistan's 2022 floods covered about a third of the country.** Monsoon rains starting in June brought record totals in Sindh and Balochistan. The government reported around 1,700 deaths, and tens of millions of people were affected.
10. **Storm Daniel's floods destroyed parts of Derna, Libya.** In September 2023, two dams upstream failed after extreme rainfall and swept through the city. Thousands died.
11. **An atmospheric river can carry a Mississippi's worth of water.** NOAA says the average one transports an amount of water vapour equal to 7.5 to 15 times the average flow at the mouth of the Mississippi River.

12. **Atmospheric rivers now have a rating scale.** In 2019, Scripps scientists introduced a five-category scale, from beneficial to hazardous, based on how much moisture they carry and how long they last.
13. **The 1927 Mississippi flood inundated about 27,000 square miles.** The Great Mississippi Flood displaced hundreds of thousands of people and led to federal flood-control legislation.
14. **Mount Wai‘ale‘ale on Kauai receives more than 400 inches a year.** It's one of the wettest spots on Earth, soaked by trade winds that hit the mountain from the northeast.
15. **Flash floods are the biggest weather killers in some years.** In the United States, they've often ranked among the top causes of weather-related fatalities, and a large fraction of the deaths occur in vehicles driven into water.
16. **Six inches of moving water can knock you down.** A foot can float a car. The National Weather Service uses the slogan "Turn Around, Don't Drown."
17. **Floods can make desert lakes appear.** After exceptional rain, Australia's Lake Eyre fills and draws birds from across the continent. The basin's rivers are far from the sea, and it can take months for water to arrive.
18. **A 2015 Atacama deluge hit the driest desert.** In March 2015, heavy rain in northern Chile caused floods and mudslides in towns that normally see almost no rain.
19. **A monsoon is a seasonal wind reversal.** The word comes from the Arabic *mawsim*, meaning season. South Asia gets most of its yearly rain in a few months.

20. **Cameras caught the Zhengzhou subway.** Video from inside train cars showed water at passengers' chests, and the clips triggered a national debate about urban drainage and preparedness.

Chapter 15: Derechos, Downbursts, and Winds That Flatten Forests

Wind is hard to film, because you only see what it moves. These are the events where it moved a lot: grain bins, forests, aircraft, and one very fast jumbo jet.

1. **The strongest wind gust ever measured outside a tornado was 408 km/h.** It hit Barrow Island, off Western Australia, during Severe Tropical Cyclone Olivia on 10 April 1996. The Bureau of Meteorology flagged the reading as suspect at first, and the WMO ratified it in 2010 after checking the instrument.
2. **For decades the record belonged to a mountaintop in New Hampshire.** On 12 April 1934 the observatory on Mount Washington recorded a 231 mph (372 km/h) gust, which stood as the world record for a surface wind until Olivia.
3. **A derecho is a long-lived windstorm that moves in a straight line.** The word is Spanish for "straight." Gustavus Hinrichs, a University of Iowa professor, proposed it in 1888. NOAA's Storm Prediction Center describes one as a swath of damage at least 240 miles long with gusts of 58 mph or more.
4. **Iowa got a derecho in August 2020.** On 10 August a line of storms crossed the state and parts of the Midwest, with wind gusts of more than 100 mph in places. Damage was estimated at over \$10 billion, one of the costliest thunderstorm events in U.S. history.

5. **A microburst is a column of sinking air that hits the ground and spreads out.** Ted Fujita identified them after the crash of Eastern Air Lines Flight 66 at New York's JFK Airport in 1975, which killed 112 people. His analysis showed that the plane had flown into a downdraft.
6. **Another microburst crash led to better detection.** Delta Flight 191 hit the ground near Dallas–Fort Worth in 1985 and 137 people died. The accident was a major push behind Doppler weather radar at airports.
7. **Supercells got their name in 1962.** Keith Browning and Frank Ludlam described a storm that hit near Wokingham, England, with a persistent rotating updraft, and called it a "supercell."
8. **Storm Eunice set England's gust record in 2022.** On 18 February, a gust of 122 mph (196 km/h) was recorded at The Needles on the Isle of Wight, the strongest ever measured in England.
9. **A "sting jet" is a narrow stream of very fast air inside a storm.** Browning proposed it in 2004 after studying the Great Storm of October 1987 in Britain, which toppled an estimated 15 million trees. Some scientists regard its role in that storm as still debated.
10. **Storm Ciara pushed a Boeing 747 to 825 mph.** On 8 February 2020, a British Airways flight from New York to London rode a jet stream of roughly 260 mph and arrived in 4 hours 56 minutes, a record for a subsonic airliner. The airspeed stayed below the speed of sound.
11. **A bomb cyclone is a real technical term.** Fred Sanders and John Gyakum at MIT coined "bomb" in 1980 for a mid-latitude storm whose pressure falls at least 24 millibars in 24 hours, scaled for latitude.

12. **Cape Denison in Antarctica is among the windiest places at sea level.** Douglas Mawson's 1911–14 expedition camped there and titled his book *The Home of the Blizzard*. Cold, dense air pours off the ice sheet nearly continuously.
13. **The Bora is a winter wind that hits the Adriatic coast.** Cold air spills over the Dinaric Alps and rushes down toward the sea, sometimes producing gusts of 150 km/h or more. Bridges and roads in Croatia close when it blows.
14. **The Santa Ana winds feed California's worst fires.** Dry air from the interior flows downhill toward the coast and warms as it goes, with gusts that can top 70 mph in the mountain passes.
15. **The mistral blows down the Rhône valley into southern France.** It's cold, dry, and can last for days. Vincent van Gogh complained about it while painting in Arles.
16. **Downslope winds heat air as they descend.** The chinook in the Rockies, the foehn in the Alps, and the Santa Ana are all examples. Air warms by compression as it sinks, which is how Loma, Montana, got a 103-degree swing in a day (see Chapter 16).
17. **A wind turbine can be shut down by too much of it.** Most large turbines stop when the wind exceeds about 25 metres per second, around 56 mph, to prevent damage.
18. **The cup anemometer dates to 1846.** John Thomas Romney Robinson of Armagh Observatory built the first one, with four hemispherical cups on a spinning arm. He assumed the cups turned at one-third of the wind's speed, and later experiments showed the true ratio was closer to one-half, so early readings ran high.

19. **Wind speeds are reported in different ways.** A sustained wind is an average over a set time, usually one or ten minutes. A gust is a brief peak, normally lasting a few seconds. The distinction explains why different agencies give different numbers for the same storm.
20. **A Fujita-scale rating is based on damage, not measurements.** Because tornadoes seldom hit an instrument, surveyors estimate wind from how buildings and trees failed.

Chapter 16: Scorching, Freezing, and Everything in Between

Temperature records come with footnotes. The thermometer matters, the site matters, and sometimes the argument runs a hundred years. These are the numbers the WMO keeps on file, plus a few things that happen when it gets too hot or too cold.

1. **The hottest air temperature on record is 56.7°C (134.1°F).** It was measured at Furnace Creek, Death Valley, on 10 July 1913. Some researchers have questioned the reading, citing possible observer error, so the record is **[DISPUTED]**, even though it's the WMO's official figure.
2. **The most reliable modern reading is 54.4°C (129.9°F).** Death Valley reached that on 16 August 2020, and again in July 2021. Many climatologists treat those as the highest well-documented temperatures.
3. **The coldest air temperature on record is -89.2°C (-128.6°F).** It was recorded at Vostok Station in Antarctica on 21 July 1983.
4. **Satellites have seen colder.** Instruments on orbiting satellites measured surface temperatures near -93°C on the East Antarctic Plateau in 2010. That's a ground-surface reading, not an air temperature, so it isn't counted in the WMO record.
5. **Oymyakon and Verkhoyansk dispute the title of coldest inhabited place.** Siberian readings of about -67°C in the early twentieth century and 1892 respectively are cited for each, and the numbers are contested.

6. **Canada hit 49.6°C in June 2021.** The temperature in Lytton, British Columbia, on 29 June broke the national record by almost five degrees. The next day a wildfire destroyed most of the village.
7. **Britain topped 40°C for the first time on 19 July 2022.** Coningsby, Lincolnshire, recorded 40.3°C. Runways at Luton Airport were damaged by the heat, and rail speeds were cut across the country.
8. **Siberia reached 38°C inside the Arctic Circle.** The reading at Verkhoyansk on 20 June 2020 was later verified by the WMO as the highest temperature recorded north of the Arctic Circle.
9. **Loma, Montana, saw a 103-degree swing in 24 hours.** On 14 and 15 January 1972, the temperature rose from -54°F to 49°F as a chinook wind arrived, the largest 24-hour change ever recorded.
10. **The 2003 European heat wave killed tens of thousands.** A study estimated about 70,000 excess deaths across Europe that summer, with France among the hardest hit.
11. **Texas froze in February 2021.** Winter Storm Uri knocked out power to more than 4 million customers. Hundreds died, and burst pipes flooded homes.
12. **Iguanas fall out of trees in cold snaps.** Florida's iguanas are cold-blooded and go into a stunned state when it drops below about 5°C. Photos of them lying on the ground have been published in 2010, 2018, and 2021, and most revive when they warm up.
13. **Dallol, Ethiopia, has the highest annual average temperature of any inhabited place.** The yearly mean is about 34°C. The area's salt flats and acid pools look like another planet.

14. **Wind chill was invented in Antarctica.** Paul Siple and Charles Passel developed the concept in 1939 during research on how fast water freezes in cold wind. The U.S. formula was revised in 2001.
15. **Frying an egg on the sidewalk is mostly a [MYTH].** Eggs set at around 62 to 70°C. A sunny pavement can reach 60°C or more, but it conducts heat poorly and the results are usually unappetising.
16. **Heat can bend railways.** Steel rails expand and can buckle, called "sun kinks." Railways in Britain and the U.S. apply speed limits during extreme heat.
17. **Heat waves are the leading weather killer in many countries.** In the United States, extreme heat has tended to cause more deaths per year than hurricanes, tornadoes, or floods.
18. **The "wet-bulb" temperature measures how well sweat cools you.** Researchers say a sustained wet-bulb temperature of about 35°C would be unsurvivable for healthy people for long. That's a **[THEORY]** based on physiology models, and real-world limits may be lower.
19. **Sound carries farther on cold, still nights.** Under a temperature inversion, with cold air near the ground and warmer air above, sound waves bend back down, so distant trains and traffic seem close.
20. **Frost fairs once took place on the Thames.** In cold winters from the 1600s to 1814, the river in London froze hard enough to hold markets. The period is often tied to the Little Ice Age, and the old London Bridge, whose narrow arches slowed the water, is often cited as a contributing factor.

Chapter 17: Rogue Waves, Meteotsunamis, and Seas Gone Mad

For centuries, sailors described waves taller than a building and were mostly told they were exaggerating. In 1995 a laser on a North Sea platform settled the argument.

1. **The Draupner wave was the first rogue wave measured by an instrument.** On 1 January 1995, a laser sensor on a gas platform in the North Sea recorded a wave of 25.6 metres (84 feet) at a time when the surrounding seas averaged about 12 metres. The platform was built to survive a 20-metre wave.
2. **Before that, rogue waves were sailors' stories.** Ship reports of walls of water had circulated for centuries, and many oceanographers assumed they were exaggerations.
3. **An ESA satellite project found a lot of them.** The MaxWave project, using radar data from ESA's ERS satellites in 2000, identified about ten waves above 25 metres in three weeks over the global oceans, though the findings are still discussed.
4. **A buoy in 2013 recorded the largest significant wave height ever.** On 4 February 2013 in the North Atlantic, between Iceland and the UK, a buoy measured a significant wave height of 19 metres, which is the average of the highest third of waves. The WMO validated it in 2016.
5. **Hurricane Ivan produced a 27.7-metre wave.** Sensors on the seabed of the Gulf of Mexico recorded a maximum individual wave of 27.7 metres (91 feet) crest to trough in September 2004, with estimates that taller ones occurred near the eyewall.

6. **A cruise ship was hit by a rogue wave in 2010.** The *Louis Majesty* was struck in the Mediterranean on 3 March 2010, off Spain, with waves breaking windows on a high deck. Two passengers died.
7. **The Agulhas Current is notorious.** Off South Africa's east coast, a strong current runs southwest while storms push waves northeast. The opposition steepens the waves, and ships have reported damage there since the 1970s.
8. **A meteotsunami is a tsunami made by weather.** A fast-moving pressure disturbance or squall line can push the sea surface and generate a wave that resembles a tsunami, though no earthquake is involved.
9. **A meteotsunami hit Nagasaki Bay in 1979.** On 31 March, a wave reportedly about 5 metres high struck, killing three people. The local name for such events is *abiki*.
10. **Daytona Beach had one in 1992.** On 3 July, a 3-metre wave reached the shore and injured about 75 people, with damage to cars on the beach, and it was later traced to a squall line over the ocean.
11. **Lake Michigan had one in 1954.** On 26 June, a sudden surge at Chicago swept anglers off piers and killed eight. It's one of the best-known meteotsunami events on the Great Lakes.
12. **Ciutadella harbour on Menorca has a local word for them.** *Rissaga* waves hit the long, narrow inlet and can surge several metres within minutes, an effect amplified by the shape of the harbour.

13. **The Edmund Fitzgerald sank on Lake Superior on 10 November 1975.** All 29 crew died in a storm that produced waves of about 10 metres. The cause remains debated, and treat the competing explanations as **[THEORY]**.
14. **El Faro was lost in Hurricane Joaquin.** The cargo ship sank on 1 October 2015 with all 33 people aboard. Its voyage data recorder, recovered from 15,000 feet down, gave investigators audio from the bridge.
15. **Sea smoke rises from freezing air over warm water.** When air well below freezing flows across open water, the water evaporates into the cold and condenses into wisps. It's common in the Arctic and over the Great Lakes in winter.
16. **The Grand Banks are among the foggiest waters in the world.** Cold Labrador Current and warm Gulf Stream meet off Newfoundland. The area sees fog on well over 100 days a year.
17. **Swells can cross an ocean.** In 1963, Walter Munk and colleagues tracked waves from Southern Ocean storms across the Pacific to Alaska, a distance of more than 10,000 kilometres.
18. **Storm surge, not wind, kills most people.** In hurricanes, the rise of sea level pushes water inland and drowns people. That's why forecasters stress surge in evacuations.
19. **Waterspouts come in two kinds.** Tornadic waterspouts form from supercell tornadoes crossing water. Fair-weather waterspouts are more common and form over warm water under growing cumulus clouds.

20. **The Beaufort scale was devised for sailors.** Francis Beaufort, a Royal Navy officer, created the scale in 1805 by describing what a ship's sails did at different wind strengths. It was later adapted to describe conditions on land.

Chapter 18: Animals Who Saw the Storm Coming

Animals notice things people can't. Some of that shows up as measurable behaviour. The rest is folklore, and this chapter tries to separate them.

1. Songbirds left Tennessee before a tornado outbreak in 2014.

Researchers led by Henry Streby tracked golden-winged warblers with tiny geolocators. The birds left their breeding grounds in late April, flew at least 1,500 kilometres, and avoided a system that spawned dozens of tornadoes and killed at least 35 people.

2. The likely cue was infrasound. The storms were still 400 to 900 kilometres away when the birds left. Some birds can register sound far below the range humans hear, and the researchers proposed that they detected the low rumble of tornadic storms. It's a **[THEORY]** supported by the evidence, not a settled fact.

3. Weather radar sees birds. When flocks of swallows or martins leave a roost at dawn, they appear on radar as rings spreading outward. Meteorologists call them roost rings, and researchers in the 1990s began using them to study populations.

4. Early radar operators called unexplained echoes "angels." Many turned out to be birds or insects. Modern dual-polarisation radar can usually tell flocks from rain.

5. BirdCast turns radar data into migration forecasts. The project, run by Cornell and partners, predicts nights when large numbers of birds will pass overhead. Cities use it to switch off lights during peak migration.

6. **Bats show up on radar too.** Millions of Brazilian free-tailed bats leave Bracken Cave in Texas on summer evenings, and the cloud is visible to nearby weather radar.
7. **Spiders fly using electricity.** In 2018, researchers Erica Morley and Daniel Robert showed that spiders can launch into the air, a behaviour called ballooning, in response to Earth's atmospheric electric field, even without wind.
8. **Darwin noticed spiders landing on his ship.** During the *Beagle* voyage in 1832, thousands of small spiders landed on the ship about 60 miles off the coast of Argentina, riding on silk threads.
9. **Rain of spiders in Australia in 2015.** In April, residents of Goulburn, New South Wales, reported thousands of spiders descending on silk threads across fields and houses. The event is consistent with mass ballooning after flooding.
10. **Bees reduce foraging before storms.** Honey bees tend to stay closer to the hive when pressure drops and humidity rises. The effect is observed, though the details of what they sense aren't fully worked out.
11. **Blacktip sharks left a Florida bay before a storm.** A 2001 tagging study in Terra Ceia Bay, near Tampa, showed that sharks moved into deeper water ahead of Hurricane Gabrielle. The researchers suggested they sensed the drop in barometric pressure.
12. **Lightning killed 323 reindeer at once.** In August 2016, a single storm struck a herd of wild reindeer on Norway's Hardangervidda plateau. The animals were clustered, and electricity spread through the ground.

13. **Cows lying down means rain is coming is a [MYTH].** Studies of cattle behaviour haven't found that it predicts rain, though herds do change posture with temperature and wind.
14. **Punxsutawney Phil isn't a good forecaster.** The Groundhog Day tradition in Pennsylvania dates back to 1887, and analyses by NOAA and others have put his accuracy somewhere around 35 to 40 percent, which is below a coin flip.
15. **Crickets chirp faster when it's warm.** Amos Dolbear described the relationship in 1897. For snowy tree crickets, count chirps in 13 seconds and add 40 for the temperature in Fahrenheit. It works only roughly, and only for that species.
16. **Toads left a pond before an earthquake in Italy.** In 2009, researchers studying a breeding colony near L'Aquila noticed most males abandoned it days before an earthquake. The link is **[UNCONFIRMED]**, because it's a single observation.
17. **Animal earthquake prediction hasn't been demonstrated.** Reports of strange pet behaviour are common after quakes, but controlled studies have not shown a reliable warning signal.
18. **Hurricanes carry birds off course.** After storms, tropical seabirds and shorebirds turn up far inland, sometimes thousands of kilometres from their usual range. Birdwatchers call these events "wrecks."
19. **Fire ants build rafts when floods hit.** The ants link their bodies into a floating mat that keeps the colony and its queen above water. Photographs of the rafts drifting through flooded Houston neighbourhoods circulated after Hurricane Harvey in 2017.

20. **Seabirds have been used as wind sensors.** In 2016, Japanese researchers published a method using GPS tags on streaked shearwaters, which soar over the sea, to estimate wind speed and direction from the birds' flight paths, which helps fill gaps in ocean observations.

Chapter 19: The Daredevils and Gadgets That Caught It All

Weather footage got better because someone built a more expensive, more sensitive, or more reckless camera. Here's some of the equipment, and some of the people behind it.

1. **The first weather satellite launched on 1 April 1960.** NASA's TIROS-1 sent back television pictures of cloud patterns, and for the first time forecasters could see storms from above.
2. **Satellites now watch continuously.** NOAA's GOES-16, launched in 2016, delivers imagery of the Western Hemisphere every few minutes and carries a lightning mapper. Japan's Himawari-8, launched in 2014, scans the full disc of Earth every 10 minutes.
3. **The first hook echo was spotted on radar in 1953.** On 9 April 1953, scientists at the Illinois State Water Survey noticed a hook shape on radar associated with a tornado near Champaign. It became a classic signature of a tornadic supercell.
4. **Hurricane Hunters have flown since 1943.** After Colonel Joseph Duckworth's first flight, the U.S. Air Force and NOAA developed regular reconnaissance missions. Today's crews fly WC-130J aircraft and NOAA P-3 Orions through storms.
5. **Dropsondes measure the inside of a hurricane.** Aircraft release small instrument packages on parachutes that report wind, pressure, temperature, and humidity as they fall to the sea.

6. **Oklahoma City's KWTW was early to buy Doppler radar.** In 1981, the station became the first commercial broadcaster to install a Doppler weather radar. Chief meteorologist Gary England used it on air for decades.
7. **The national Doppler network came in the 1990s.** The WSR-88D (NEXRAD) system comprises roughly 160 radars across the United States. It replaced older equipment and improved tornado warning times.
8. **Doppler on Wheels is a radar on a truck.** Josh Wurman started the program in 1995. One of its trucks measured the 301 mph wind in the 1999 Moore tornado, and the fleet took part in the VORTEX2 project.
9. **TOTO was a barrel with sensors.** The TOTOtable Tornado Observatory was developed in the 1980s to be placed in a tornado's path. It never got a direct hit, though the movie *Twister* (1996) borrowed the idea for its own sensor package.
10. **The word "tornado" was once banned from forecasts.** In the late 1880s the Army Signal Service stopped using it, fearing panic, and the ban lasted for decades. On 25 March 1948, Air Force forecasters Ernest Fawbush and Robert Miller at Tinker Air Force Base issued a tornado forecast that verified, and it helped change official attitudes.
11. **David Hoadley has chased since 1956.** He's often called a pioneer of modern storm chasing and founded the magazine *Storm Track* in 1977, which connected chasers and researchers.
12. **SKYWARN trained spotters across the country.** The National Weather Service started the program in the 1960s. Hundreds of thousands of volunteers now report tornadoes, hail, and floods.

13. **The Tornado Intercept Vehicle is an armoured truck for filming.** Filmmaker Sean Casey built the first one in the early 2000s to put IMAX cameras close to tornadoes. Its successor has hydraulic spikes that anchor it to the ground, and it was featured in the 2011 film *Tornado Alley*.
14. **Doorbell cameras and dashcams changed the footage.** Many of the recent famous clips of tornadoes, hail, and floods were shot by ordinary devices left running by their owners. Researchers now collect them as data.
15. **Smartphones have barometers.** Some phones record air pressure, and researchers have explored collecting the readings from thousands of devices to improve short-term forecasts.
16. **The public can report weather by phone.** The National Severe Storms Laboratory's mPING app, launched in 2012, lets anyone report hail, rain type, or ice, and the reports feed research and radar verification.
17. **The Enhanced Fujita scale started in 2007.** It replaced the original F-scale with a damage-based system using 28 types of damage indicators, from houses to trees.
18. **A TV crew's SUV was thrown by the El Reno tornado.** On 31 May 2013, a vehicle carrying a Weather Channel crew was lifted and tossed by the storm. They survived with injuries, and the day, which also killed three experienced chasers, led to widespread discussion of chaser safety.
19. **High-speed cameras show lightning in slow motion.** Phantom cameras record at hundreds of thousands of frames per second. Footage of leader channels branching toward the ground has helped researchers study how a strike chooses its path.

20. **Time-lapse made supercells famous.** Photographers such as Mike Olbinski have produced time-lapse films of Plains storms in the 2010s that millions have watched, helping the public see storm structure.

Chapter 20: Weather on Other Worlds

Earth's weather runs on water. Elsewhere in the solar system it runs on methane, sulfuric acid, carbon dioxide, and a few things nobody has seen directly. Spacecraft cameras have sent back the proof, and telescopes are starting to read the skies of planets around other stars.

1. **Jupiter's Great Red Spot is a storm wider than Earth.** It's an anticyclone that has been observed for at least 150 years, and it has been shrinking. It rotates counterclockwise, completing a turn in roughly six days, and winds along its edge exceed 400 km/h.
2. **Juno flew over it in 2017.** On 10 July 2017, NASA's Juno spacecraft passed about 9,000 kilometres above the storm's cloud tops and returned close-up images.
3. **Saturn has a six-sided jet stream.** The hexagon sits around the north pole, roughly 30,000 kilometres across, with each side about 14,500 kilometres long. Voyager 1 found it in 1980 and Cassini photographed it in detail beginning in 2004. Its winds blow at about 320 km/h.
4. **Saturn gets a giant storm about every 30 years.** The Great White Spot that erupted in 2010 wrapped around the planet and was tracked by Cassini.
5. **Venus has clouds of sulfuric acid.** The planet's surface temperature is about 465°C, and acid droplets in the clouds evaporate well above the ground, so the rain never lands.

6. **Venus's atmosphere rotates faster than the planet.** The cloud tops circle the planet in about four Earth days, while Venus takes 243 days to rotate once on its axis.
7. **Lightning on Venus is debated.** Past missions gave conflicting hints. In 2021, NASA's Parker Solar Probe detected radio whistler waves during a flyby that were consistent with lightning. The claim remains **[UNCONFIRMED]** until a direct optical observation.
8. **Jupiter has lightning, first seen by Voyager 1 in 1979.** Juno has since recorded it in detail, including flashes at the poles.
9. **Juno's data point to "mushballs."** Researchers proposed in 2020 that ammonia and water combine to form slushy hail deep in Jupiter's atmosphere. It's a **[THEORY]** based on indirect measurements.
10. **Mars has dust devils and global dust storms.** Rovers have photographed dust devils crossing the plains, and a storm in 2018 covered the whole planet (see Chapter 11).
11. **Mars has snow, but it evaporates before it lands.** NASA's Phoenix lander's laser radar saw ice crystals falling from clouds in 2008 that vaporised before reaching the ground.
12. **Titan has methane rain.** Saturn's largest moon has clouds, rivers, and lakes of liquid methane and ethane. The Huygens probe landed there on 14 January 2005, and Cassini mapped the lakes.
13. **Neptune has the fastest winds in the solar system.** Winds reach about 2,000 km/h (1,200 mph), roughly five times stronger than anything recorded on Earth. Voyager 2 measured them in 1989.

14. **Neptune's Great Dark Spot vanished.** Voyager 2 discovered it in 1989, and when the Hubble Space Telescope looked in 1994 it was gone. New dark spots have appeared since.
15. **Diamond rain on Neptune and Uranus is a [THEORY].**
Laboratory experiments in 2017 showed that extreme pressure could squeeze hydrocarbons into diamond. Whether it actually rains down inside those planets hasn't been observed.
16. **Pluto has blue haze.** NASA's New Horizons saw layers of haze rising roughly 200 kilometres above the surface in 2015, with a blue tint.
17. **An exoplanet has winds of 8,700 km/h.** Astronomers measured the flow on HD 189733b, blowing from the day side to the night side, in 2015. It was the first time weather on an extrasolar planet was mapped directly.
18. **That planet may be blue because of glass.** Hubble observations in 2013 showed HD 189733b has a blue colour, probably from silicate particles in its clouds. The idea of "glass rain" is an inference rather than an observation, so treat it as a **[THEORY]**.
19. **WASP-76b may have iron rain.** In 2020, astronomers reported iron vapour on the evening edge of the planet's day side, which they interpreted as iron condensing on the night side. That interpretation is still **[THEORY]**.
20. **The Sun has its own weather.** Coronal mass ejections, flares, and the solar wind affect every planet, as Chapter 13 describes. Earth's weather is just the small part we can feel.

Sources and Further Reading

Weather facts rest on agencies, archives, and published studies. Below are the organisations and papers behind the figures in this book. Records change, so check the issuing body for current values.

Agencies and archives

- World Meteorological Organization (WMO): World Weather & Climate Extremes Archive; International Cloud Atlas (2017 edition); 2025 certification of the 829-kilometre lightning megaflash
- National Oceanic and Atmospheric Administration (NOAA): National Hurricane Center, National Weather Service, Storm Prediction Center, Space Weather Prediction Center, National Centers for Environmental Information, Atlantic Oceanographic and Meteorological Laboratory, Pacific Marine Environmental Laboratory
- National Aeronautics and Space Administration (NASA): Earth Observatory, Jet Propulsion Laboratory, and the Cassini, Juno, New Horizons, Fermi, Voyager, and Parker Solar Probe missions
- National Severe Storms Laboratory (NSSL)
- European Space Agency (ESA): MaxWave project and ASIM instrument
- Australian Bureau of Meteorology: reports on Severe Tropical Cyclone Olivia and the 2003 Canberra fires
- National Snow and Ice Data Center (NSIDC)
- Guinness World Records (for items identified as Guinness entries in the text)

Papers and books

- Bajkov, A.D. "Do Fish Fall from the Sky?" *Science*, 1949.
- Briggs, M.S., et al. Electron–positron beams from terrestrial lightning observed with Fermi GBM. *Geophysical Research Letters*, 2011.
- Cen, J., Yuan, P., and Xue, S. Observation of the optical and spectral characteristics of ball lightning. *Physical Review Letters*, 2014.
- Grant, R.A., and Halliday, T. Predicting the unpredictable: evidence of pre-seismic anticipatory behaviour in the common toad. *Journal of Zoology*, 2010.
- Heymsfield, A.J., et al. Formation of hole-punch and canal clouds. *Science*, 2011.
- Khaykin, S., et al. The 2019/20 Australian wildfires generated a persistent smoke-charged vortex rising up to 35 km altitude. *Communications Earth & Environment*, 2020.
- Knipp, D.J., et al. Extreme Space Weather and the Vietnam War: the August 1972 event. *Space Weather*, 2018.
- Kraus, D., et al. Formation of diamonds in laser-compressed hydrocarbons at planetary interior conditions. *Nature Astronomy*, 2017.
- Loudén, T., and Wheatley, P.J. Spatially resolved eastward winds and rotation of HD 189733b. *The Astrophysical Journal Letters*, 2015.
- Morley, E.L., and Robert, D. Electric fields elicit ballooning in spiders. *Current Biology*, 2018.
- Munk, W.H., Snodgrass, F.E., and Gilbert, F. Waves across the Pacific. 1963.

- Pasko, V.P., et al. Electrical discharge from a thunderstorm top to the lower ionosphere. *Nature*, 2002.
- Streby, H.M., et al. Tornadoic storm avoidance behavior in breeding songbirds. *Current Biology*, 2015.
- Wang, D.W., et al. Extreme waves under Hurricane Ivan. *Science*, 2005.
- Yu, P., et al. Black carbon lofts wildfire smoke high into the stratosphere to form a persistent plume. *Science*, 2019.
- Howard, Luke. *On the Modification of Clouds*. 1803.
- Mawson, Douglas. *The Home of the Blizzard*. 1915.
- Pretor-Pinney, Gavin. *The Cloudspotter's Guide*. 2006.

A note on accuracy

Record figures were checked against agency or news reports of the agency announcements during writing. Facts attributed to historical accounts, folklore, or single observations are labelled as such in the text. If you find an error or a newer record, the author would like to hear about it.

Thank You for Reading

If this book made you look at the sky differently, a short review helps other curious readers find it.

About the author: [Author Name] writes about science, nature, and the odd corners of the natural world. Replace this paragraph with your own bio.